

PORTFOLIO

DATA ANALYST



PROFILE

Mohamad Thoriq Adira | Data Analyst

Bachelor of Science in Mathematics, Universitas Andalas

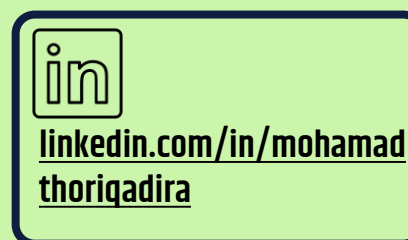
Bridging the gap between mathematical theory and **data-driven decision-making**. As a Mathematics graduate specializing in Data Analysis, I focus on translating raw numbers into visual stories that drive strategic impact. With a technical toolkit featuring **SQL**, **Python**, **Excel**, and **Power BI**, I am an adaptable fast-learner and a collaborative team player ready to contribute to dynamic, data-centric environments



SKILL & TOOLS



- **Microsoft Excel** : Capable of data profiling, basic data cleaning, Power Query transformation, Pivot Table analysis, lookup functions, and initial dataset validation.
- **SQL Server** : Capable of writing DDL and DML queries, creating database schemas, building fact and dimension tables, cleaning and transforming data, creating calculated columns, designing star schema models, and developing SQL views for reporting.
- **Python** : Capable of using Python for data processing, validation, automation, exploratory analysis, and supporting analytics workflows using libraries such as Pandas, Numpy and Matplotlib.
- **Power BI** : Developed interactive business dashboards using data modeling, DAX measures, slicers, dynamic visuals, KPI cards, and report navigation to deliver actionable insights.



BEUTY MARKETPLACE SALES ANALYSIS

Business Understanding

Beauty & Cosmetic Company operates across multiple sales channels, including third-party platforms, physical stores, and an official website. While the company is experiencing strong overall sales, the management team struggles with blind spots regarding channel profitability, unclear customer preferences, and recurring operational issues such as product returns.

The management wants to understand:

- Which sales channels and products generate the most profit (not just volume)?
- Who their core customers are and what drives their purchasing behavior?
- What are the main reasons behind product returns and order cancellations?
- How to optimize marketing budgets, inventory allocation, and fulfillment processes?

Note

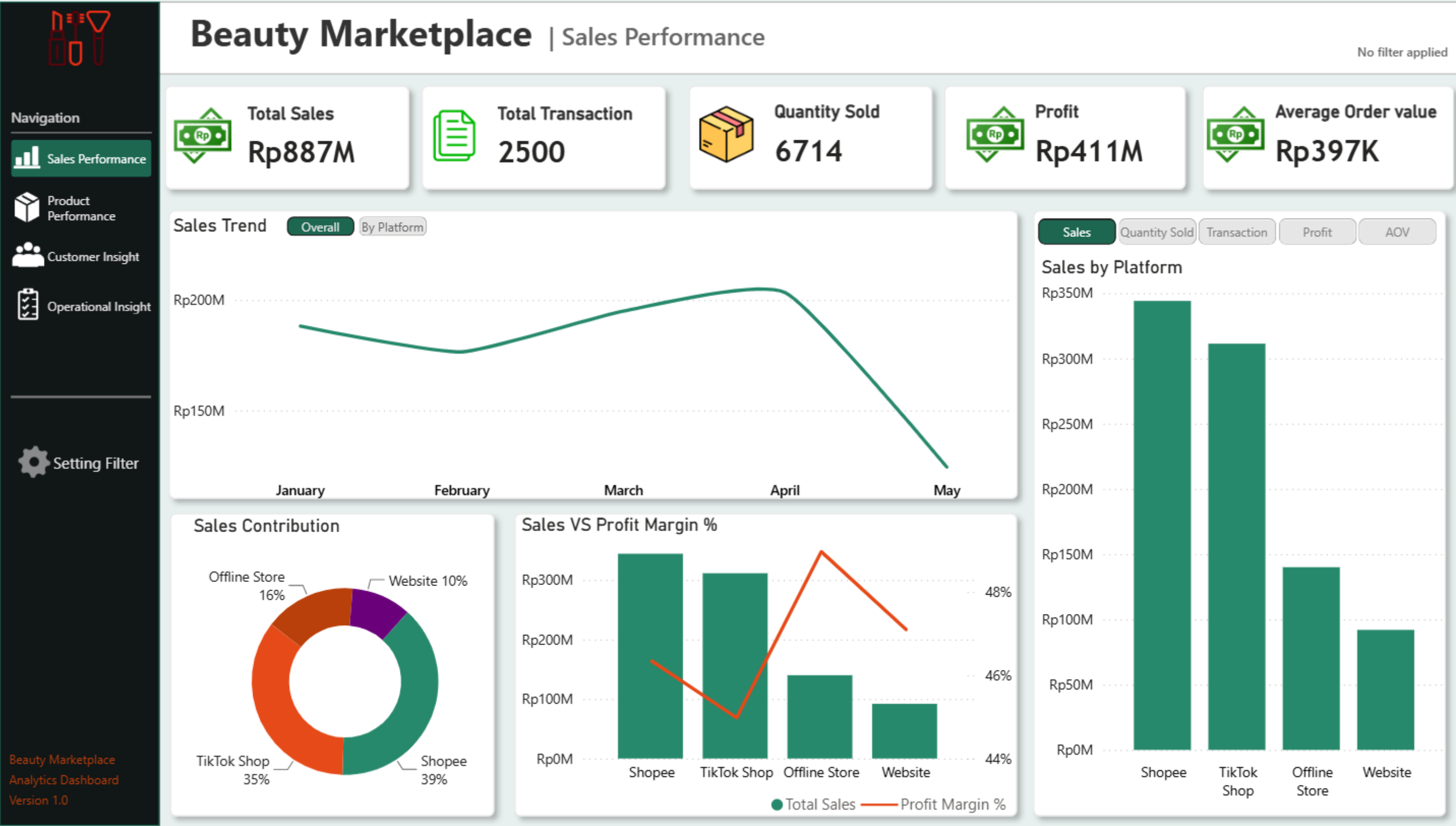
This deck presents the business-focused data story. For the complete technical workflow, including data understanding, SQL cleaning, exploratory analysis, data modeling, DAX measures, and Power BI development.

[Click here to read the full documentation on Medium article.](#)



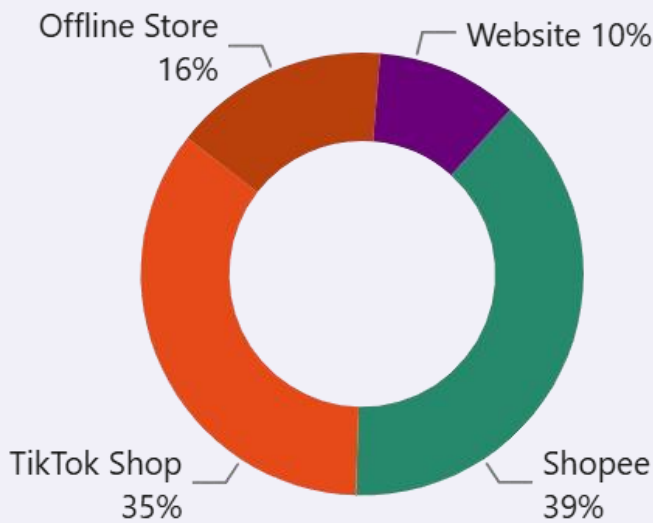
Sales Performance Dashboard

[View Interactive Dashboard](#)



Sales Performance

Sales Contribution



Sales VS Profit Margin %

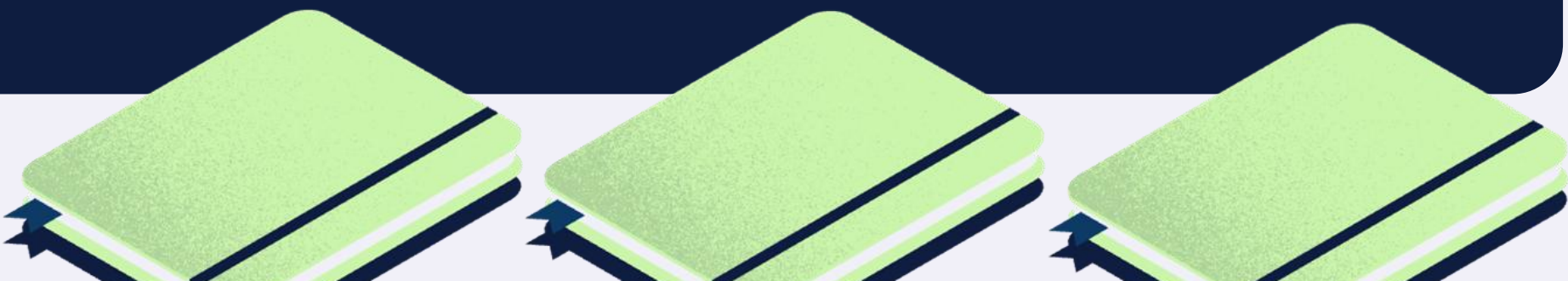


Which platforms generate the most sales and profit?

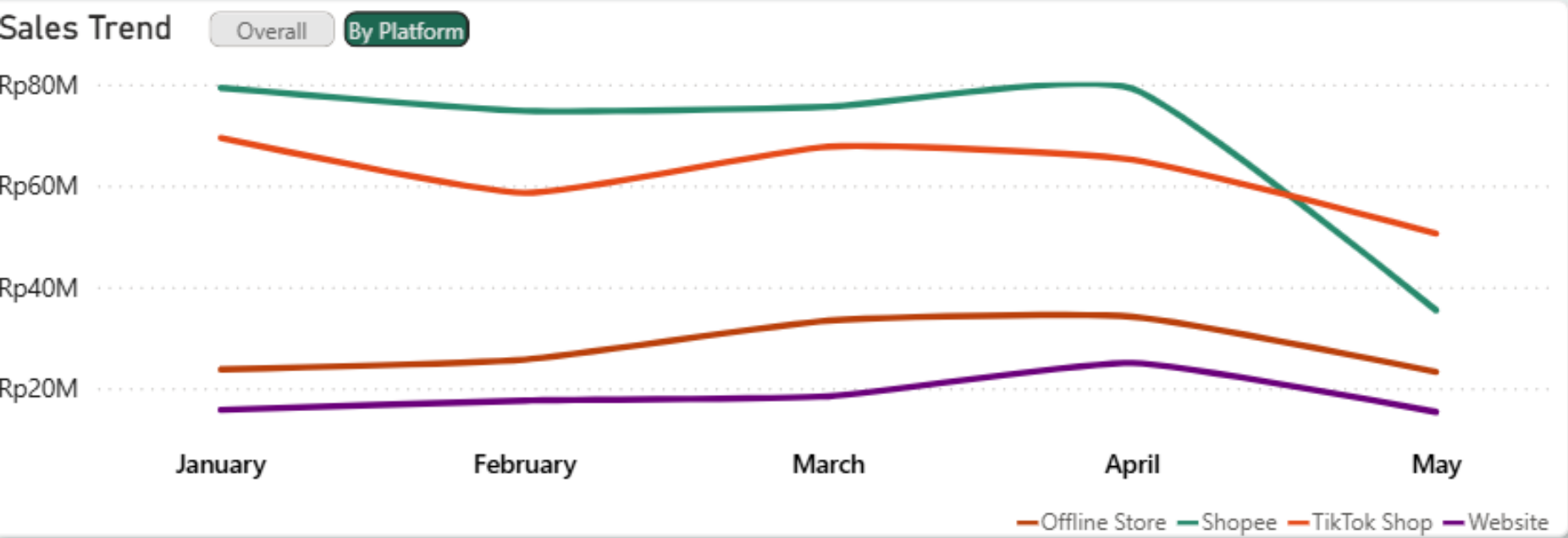
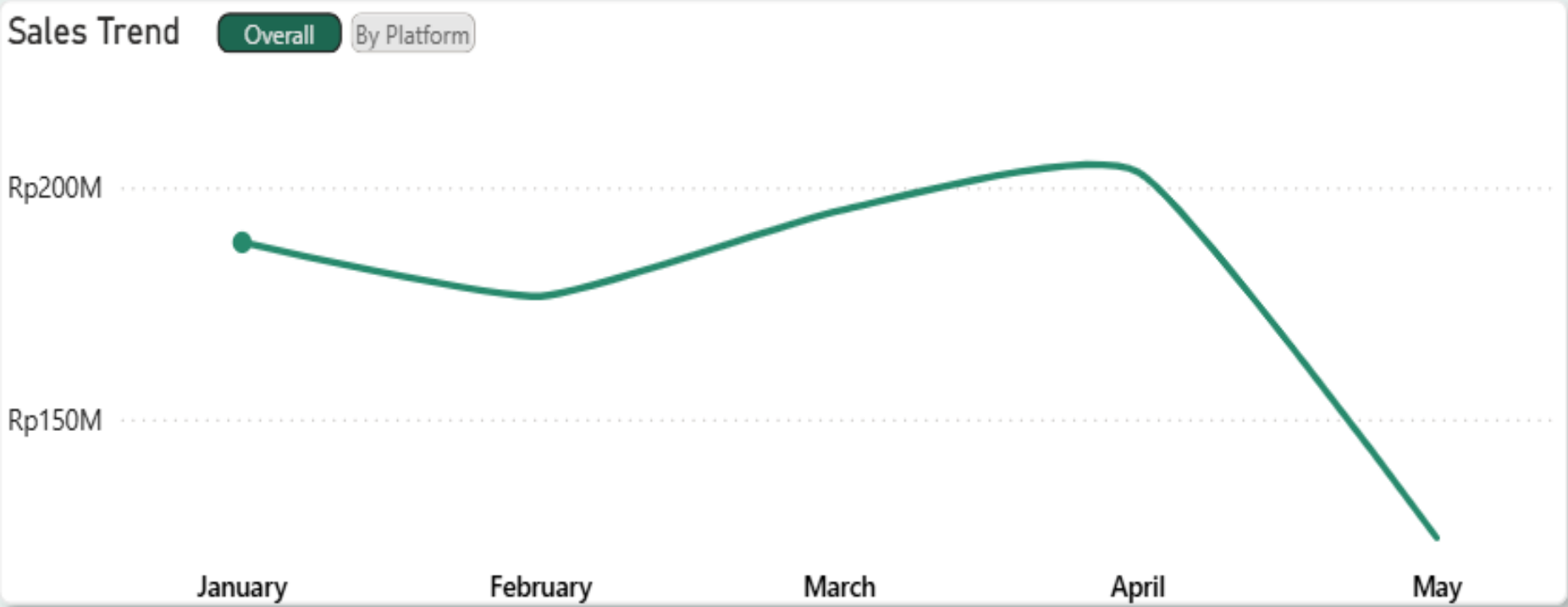
Shopee and **TikTok Shop** dominate the ecosystem, generating 74% of total sales volume. However, **TikTok Shop** yields the lowest relative profit margin (44.98%) due to heavy discounting and platform fees.

Which channel has the highest profit margin?

The **Official Website** holds the highest profit margin across the ecosystem (48%), yet it currently contributes the least to overall revenue (10%).



Sales Performance

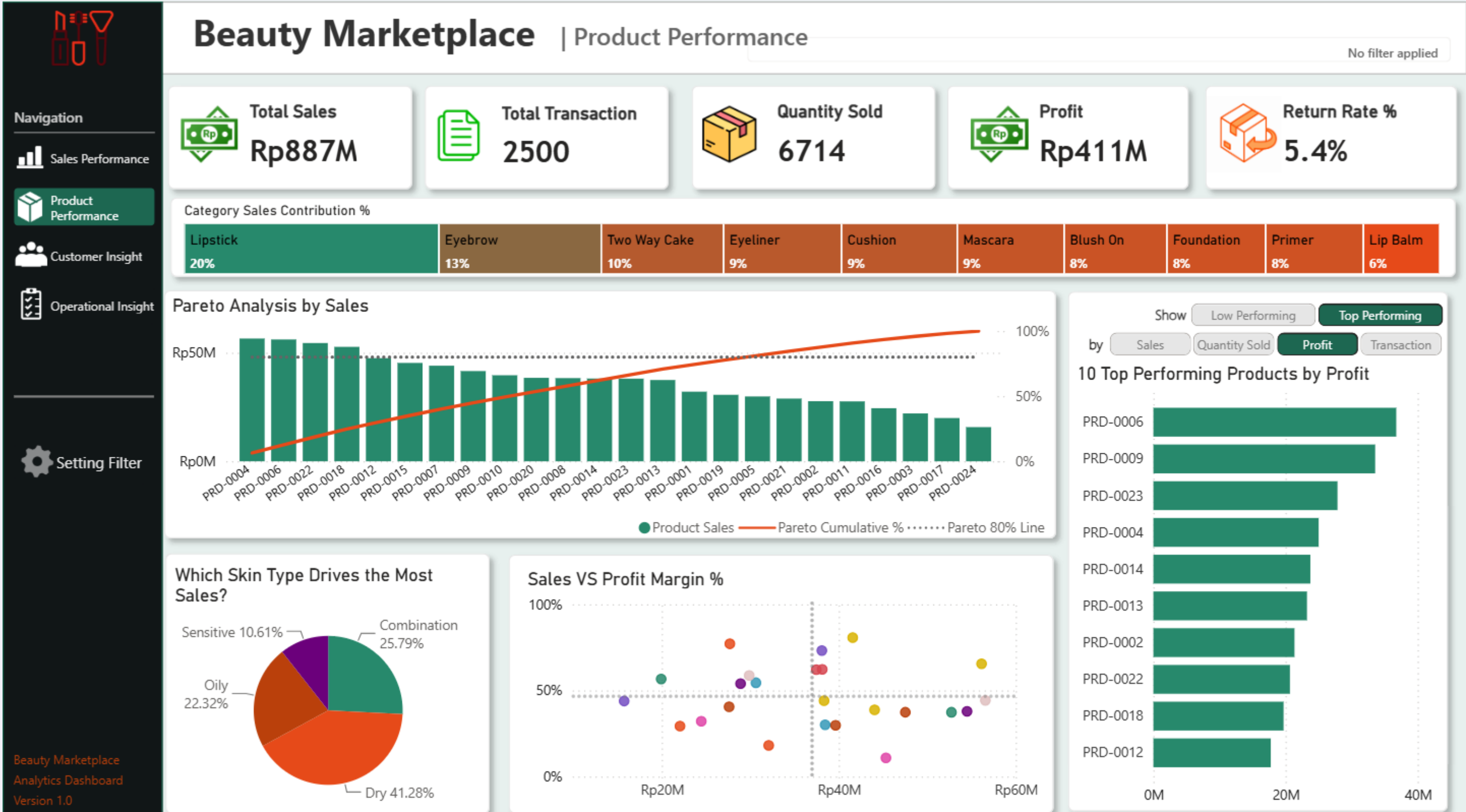


What are the overall sales trends over time?

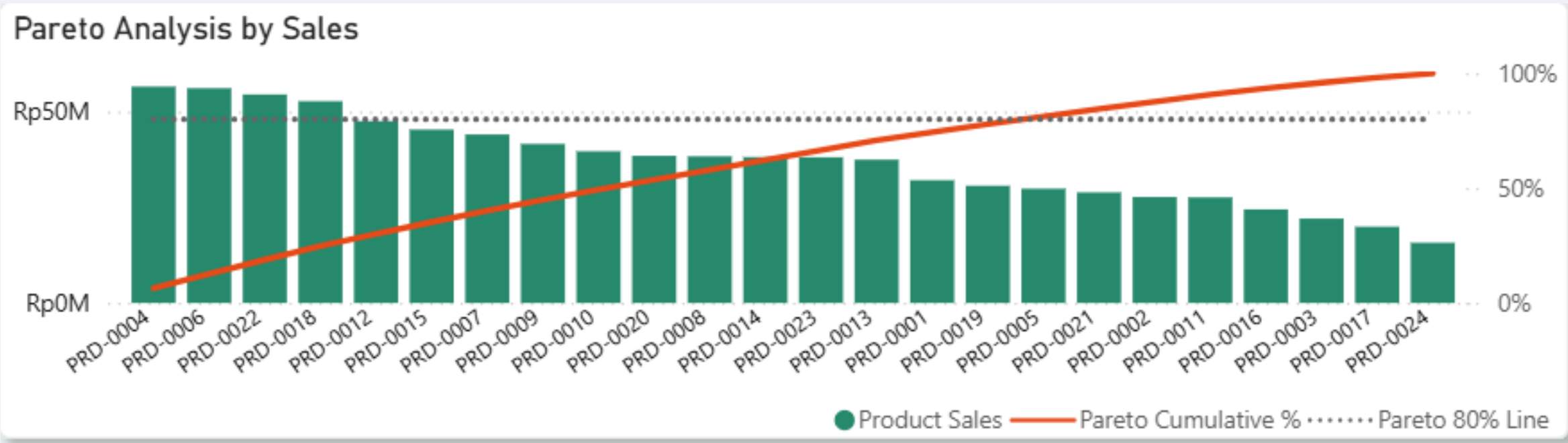
Total sales trended overall upward steadily from January, hit a peak in **April**, but experienced a steep and sudden decline in May due to incomplete data on May. However, there was a slight decline in February, mainly driven by lower sales on TikTok Shop from Rp69M to Rp58M and Shopee from Rp79M to Rp74M

Product Performance Dashboard

[View Interactive Dashboard](#)



Product Performance



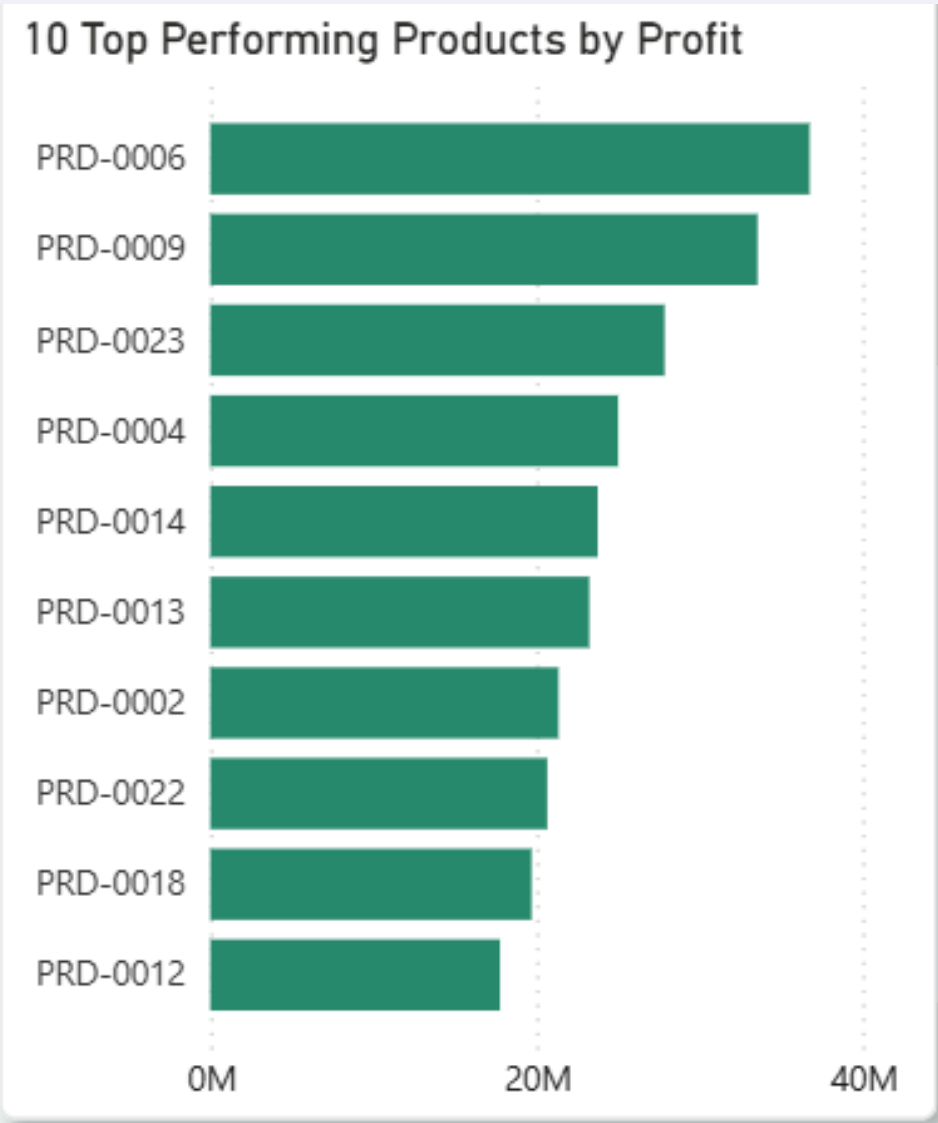
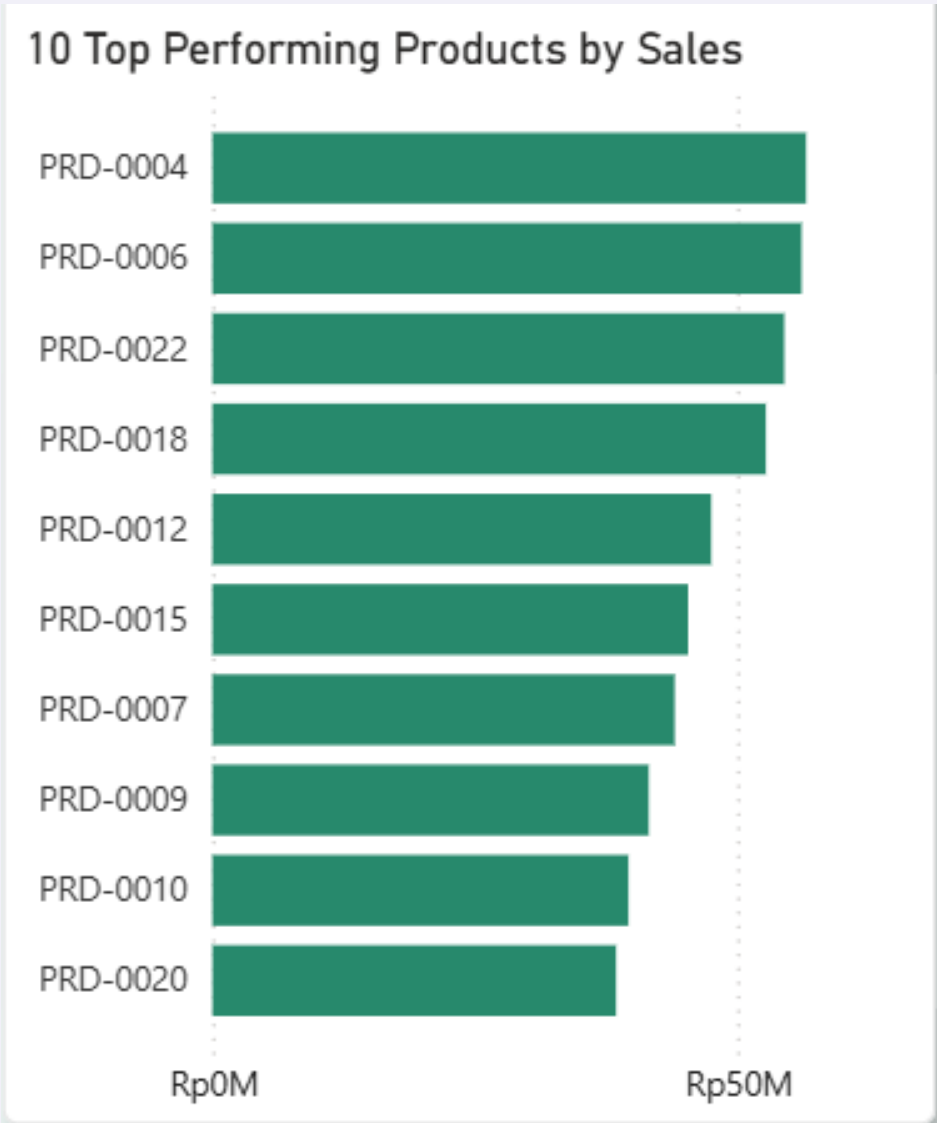
What are the best-selling products and categories?

Lipstick is the foundational revenue driver (20% contribution). At the SKU level, **Silk Two Way Cake (PRD-0004)** leads in raw sales volume.

Product Pareto Distribution

Product sales are relatively well-distributed across multiple SKUs. Although several products act as key sales drivers, the business does not rely heavily on only a few products. This indicates a balanced product portfolio, while low-performing SKUs should be reviewed for bundling, promotion, or inventory optimization.

Product Performance



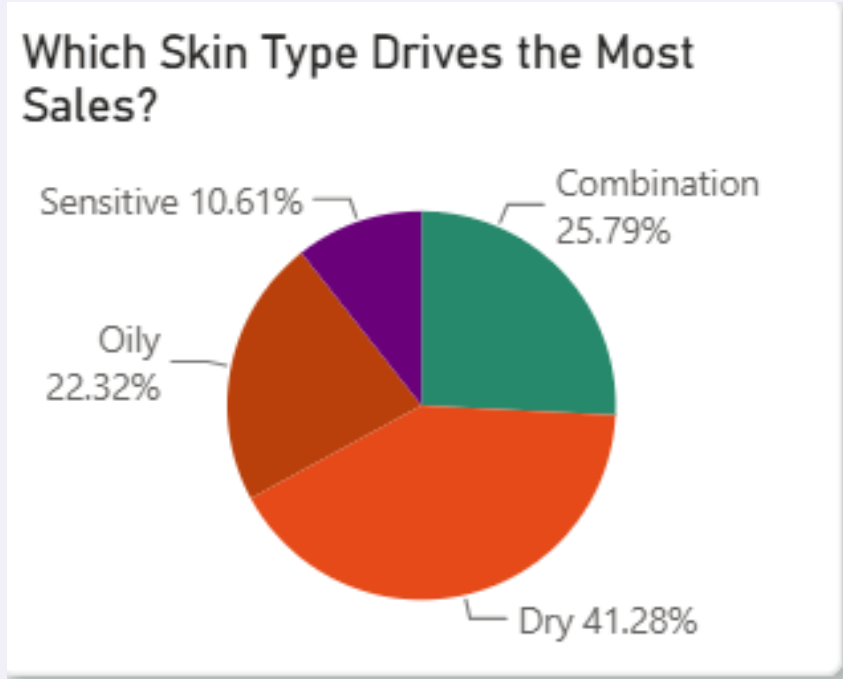
Which products contribute the most to profit?

While **Silk Two Way Cake (PRD-0004)** drives the most sales, diagnostic data indicates that other products like :

- **Glow Matte Lip Cream (PRD-0006)**
- **Lip Mousse (PRD-0009)**
- **Berry Lip Balm (PRD-0023)**

are actually more efficient at generating net profit, revealing margin dilution on top-selling items.

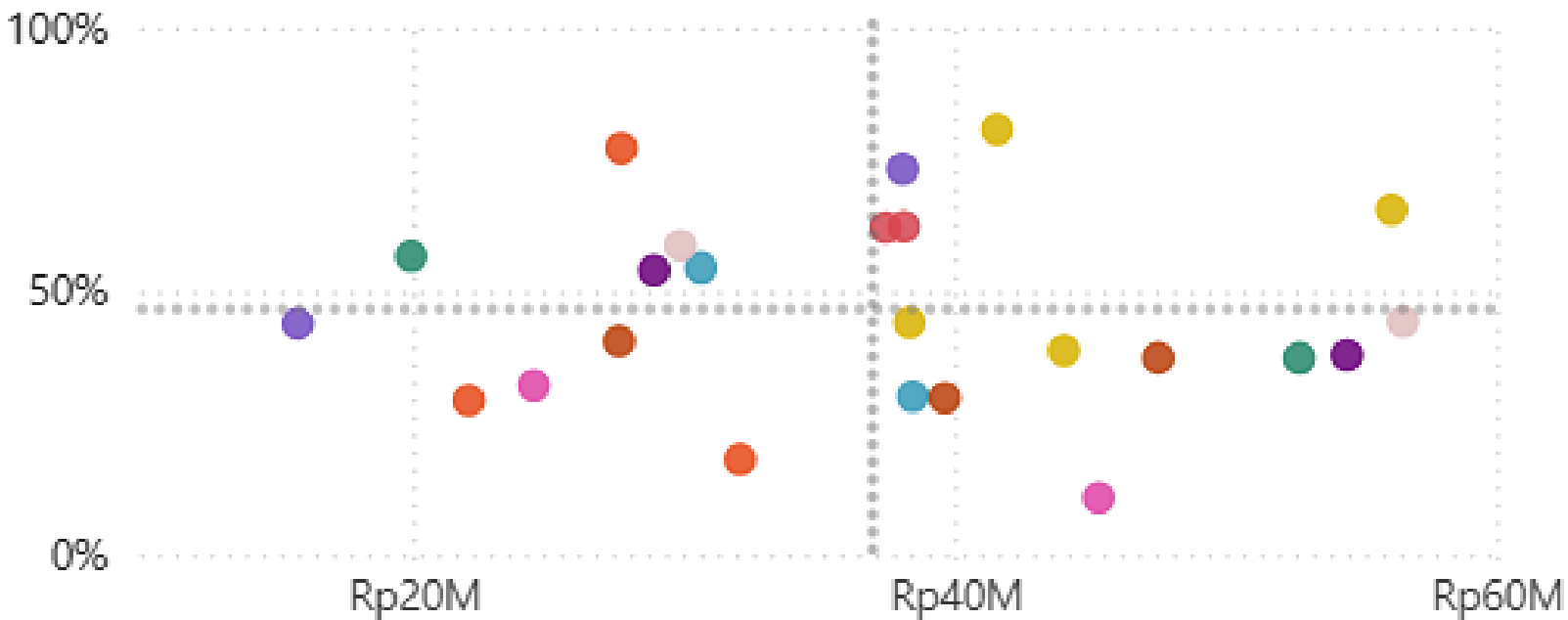
Product Performance



Which customer skin types drive the most sales?

Customers with **Dry Skin** account for a massive 41.28% of all sales, establishing a highly specific and dominant product-market fit.

Sales VS Profit Margin %

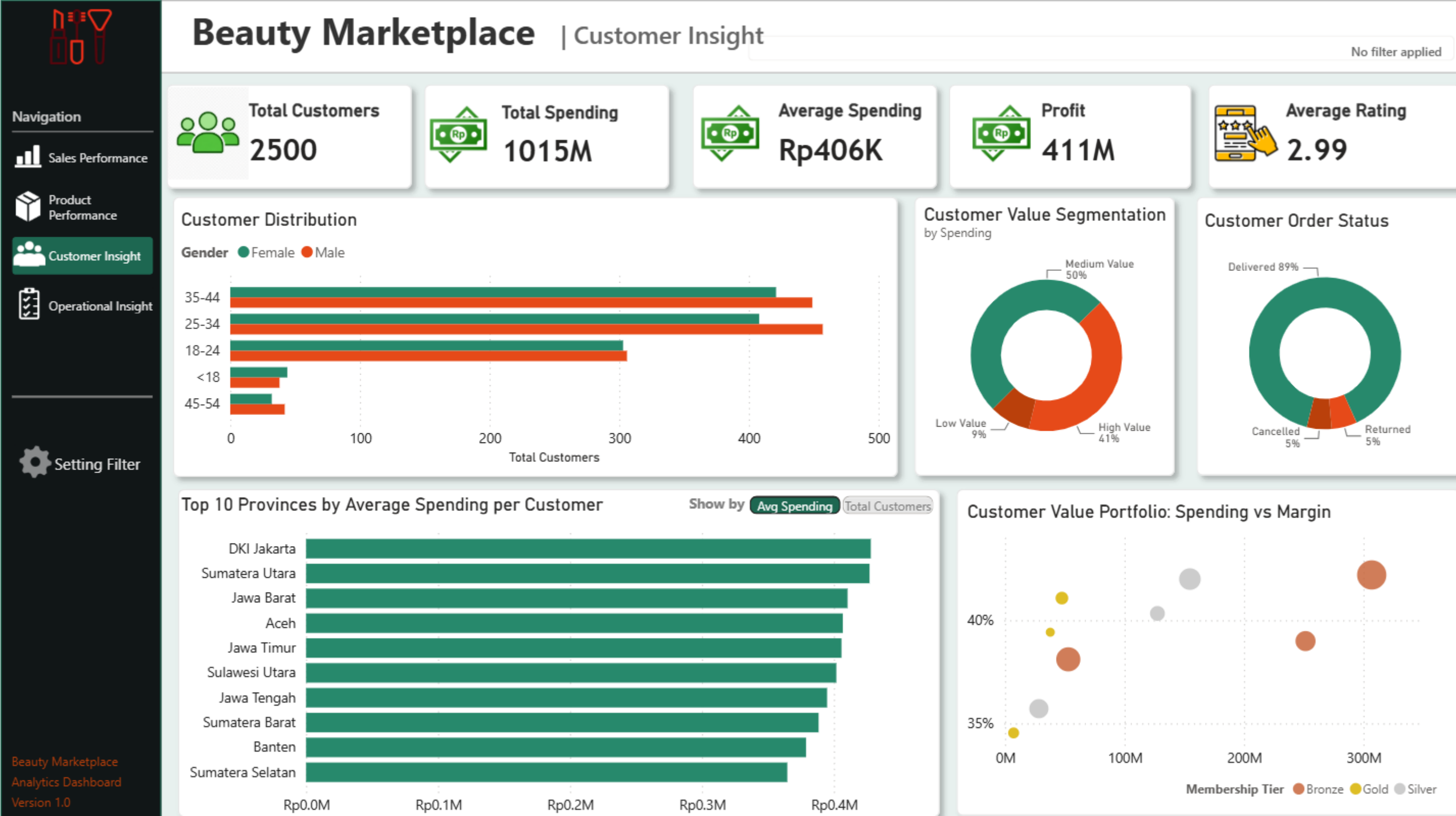


Profitability Segments

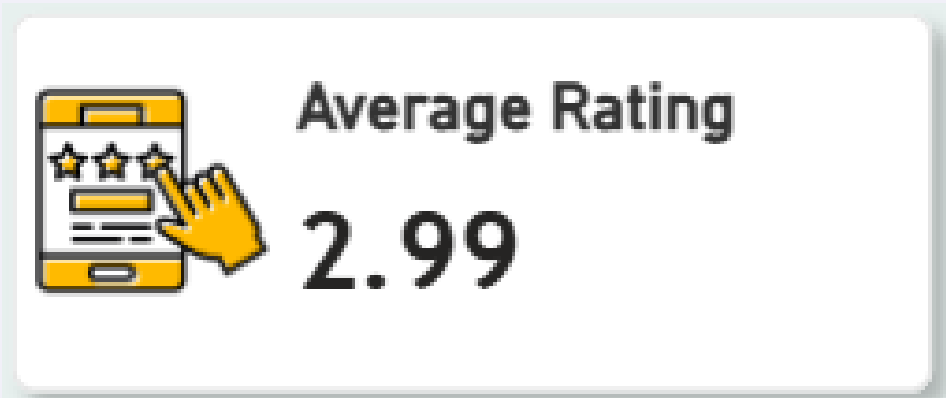
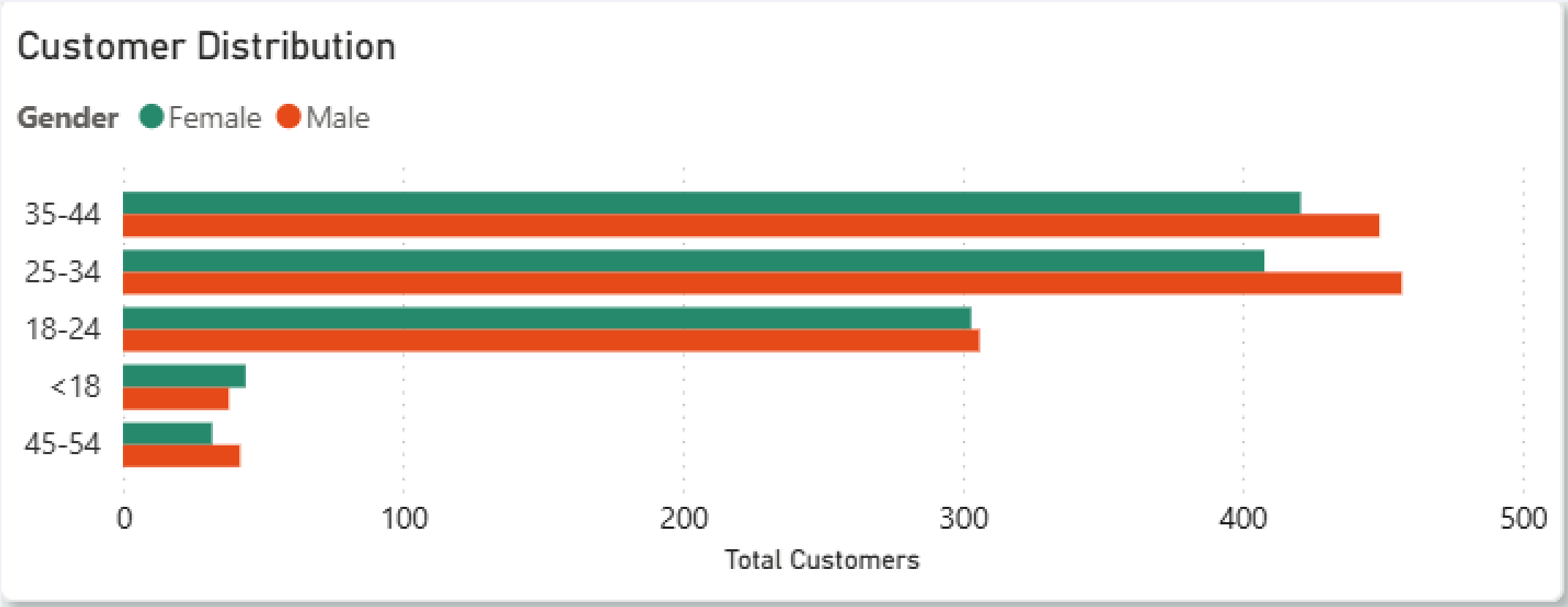
Several products show strong performance with both high sales and high profit margin. However, some high-sales products still have low margins, indicating potential profit leakage such as **Flawless Foundation (PRD-0015)**. These products should be reviewed to optimize pricing, discounts, and promotion strategy.

Customer Insight Dashboard

[View Interactive Dashboard](#)



Customer Insight



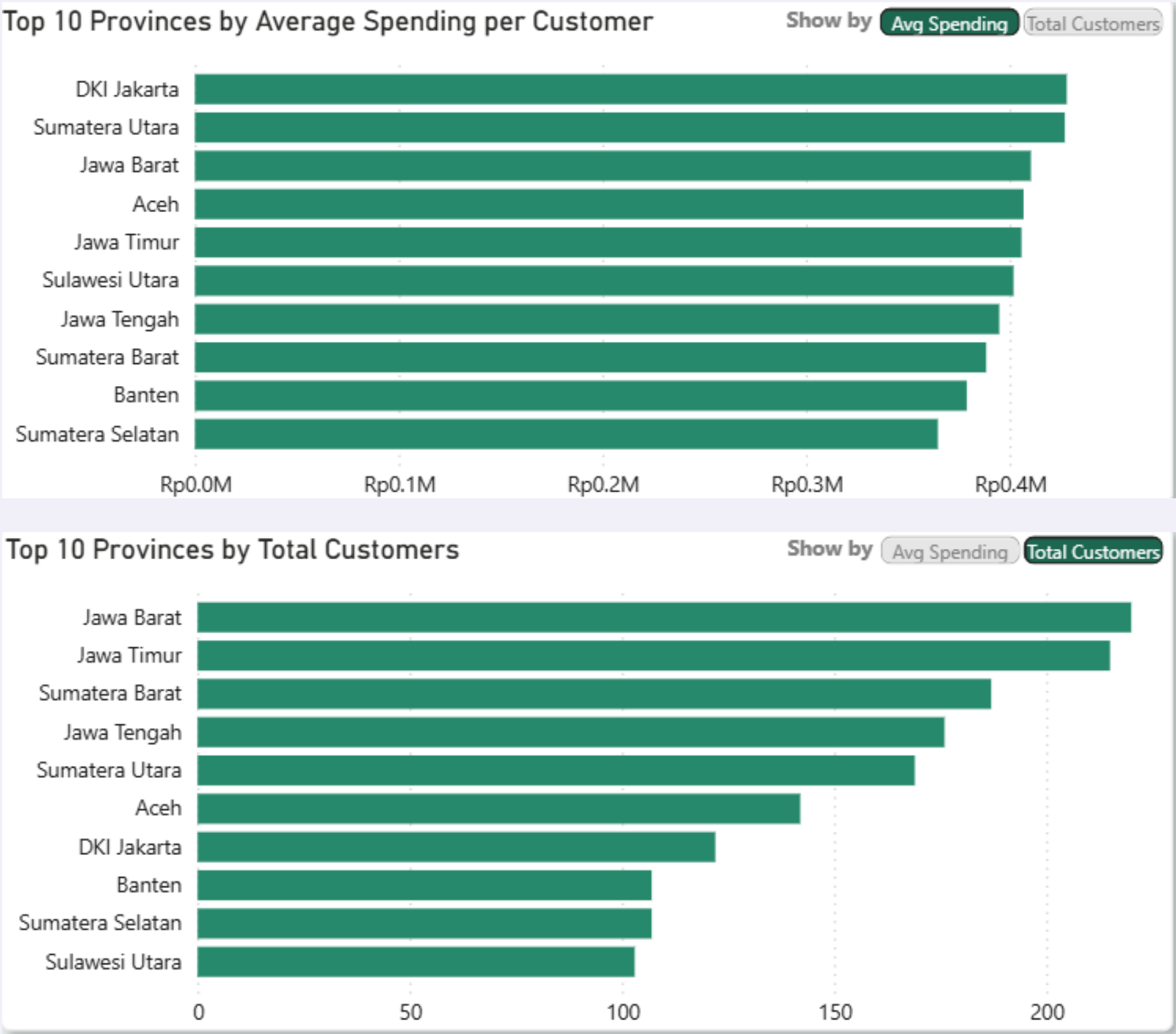
How satisfied are our customers based on their ratings?

Despite acquiring a highly valuable user base (Average Spending: Rp406K), the average rating is critically low at 2.99/5.00, signaling severe post-purchase friction.

What is the age and gender distribution of buyers?

The 25–44 age brackets completely dominate the user base. Surprisingly, Male shoppers slightly surpass Female shoppers in these core productive-age groups.

Customer Insight



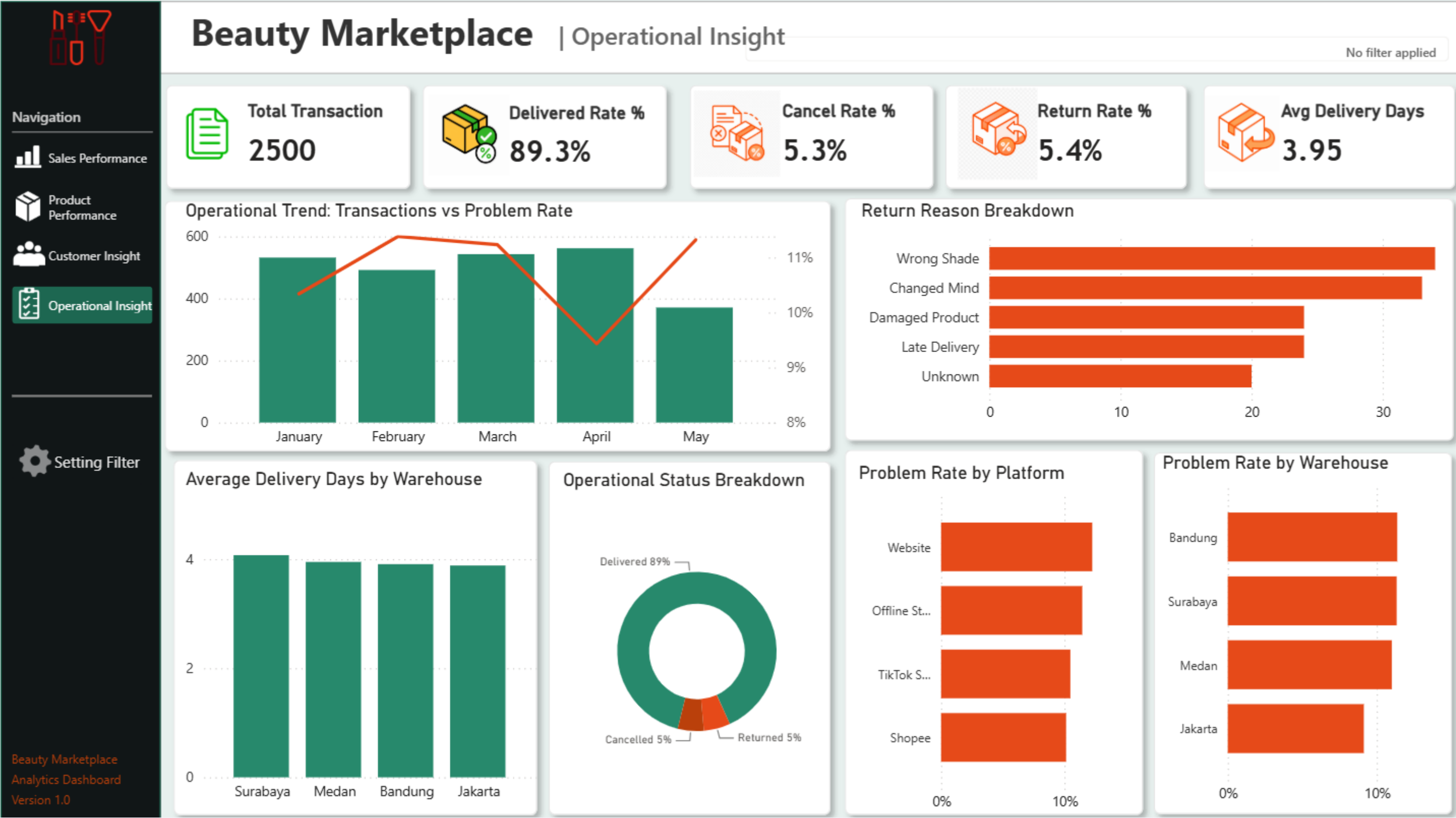
Which locations contribute the most customers and sales?

DKI Jakarta and Sumatera Utara have the highest average spending per customer, indicating strong purchasing power and high-value customer potential. Meanwhile, Jawa Barat and Jawa Timur contribute the largest number of customers, showing that these provinces are the main volume markets.

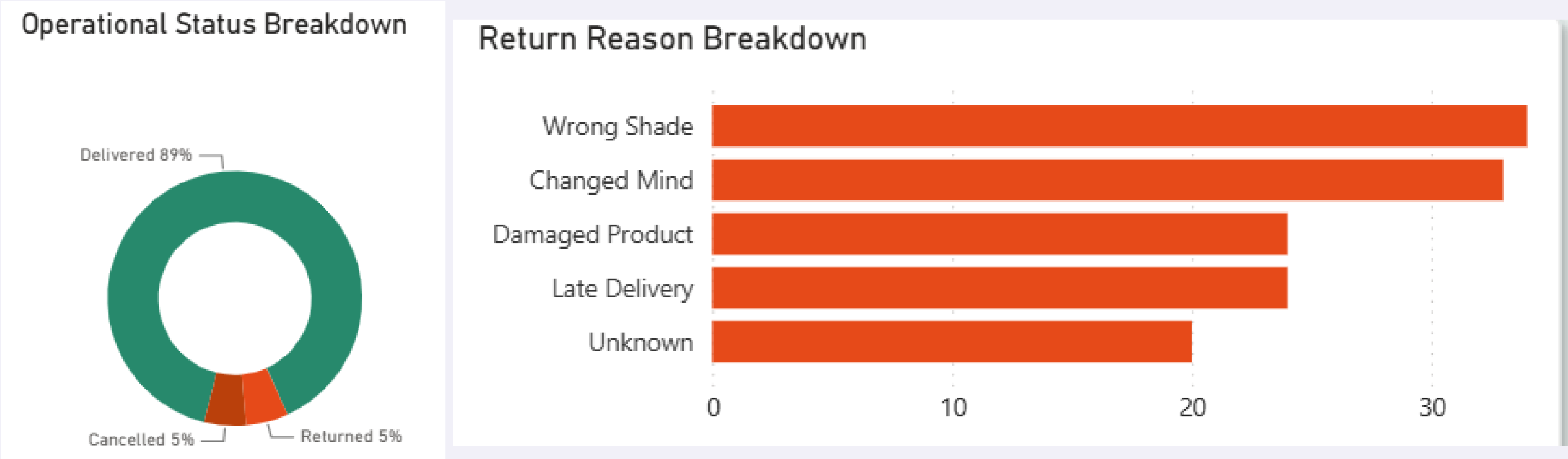
This suggests that DKI Jakarta and Sumatera Utara should be targeted for premium upselling, while Jawa Barat and Jawa Timur should be prioritized for customer acquisition, retention campaigns, and volume-based promotions.

Operational Insight Dashboard

[View Interactive Dashboard](#)



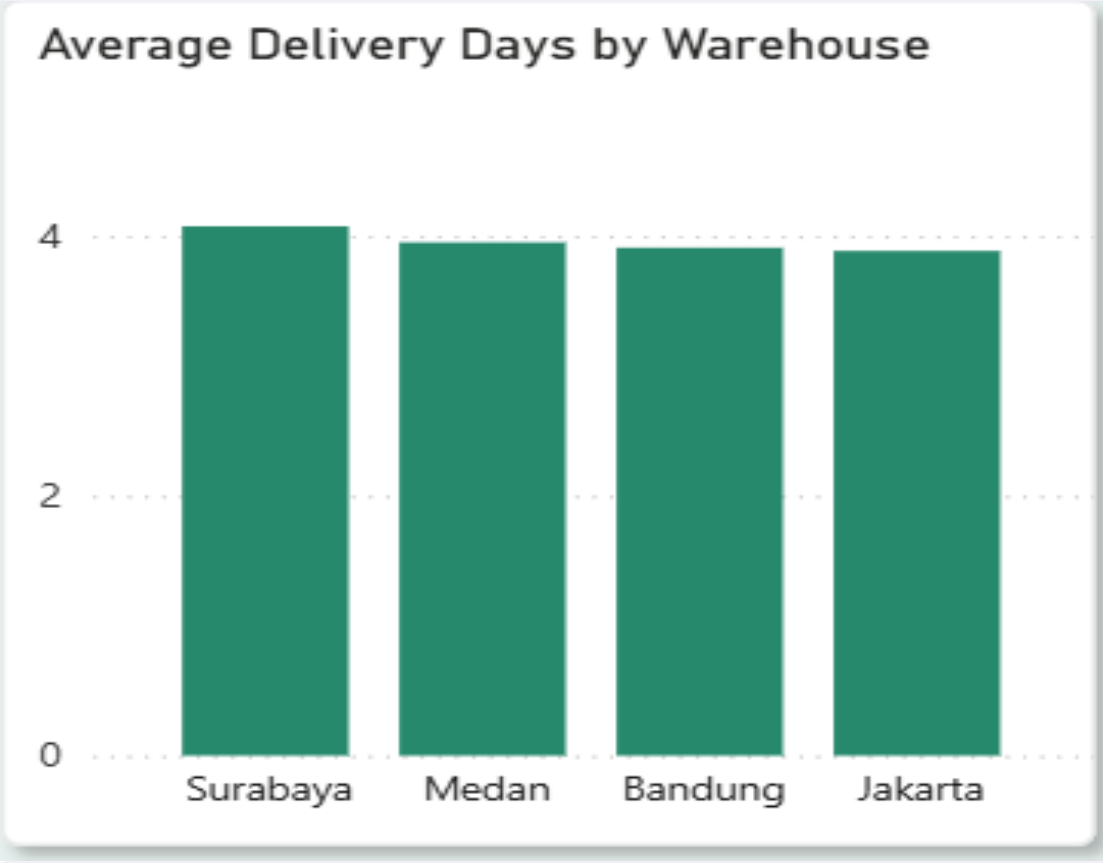
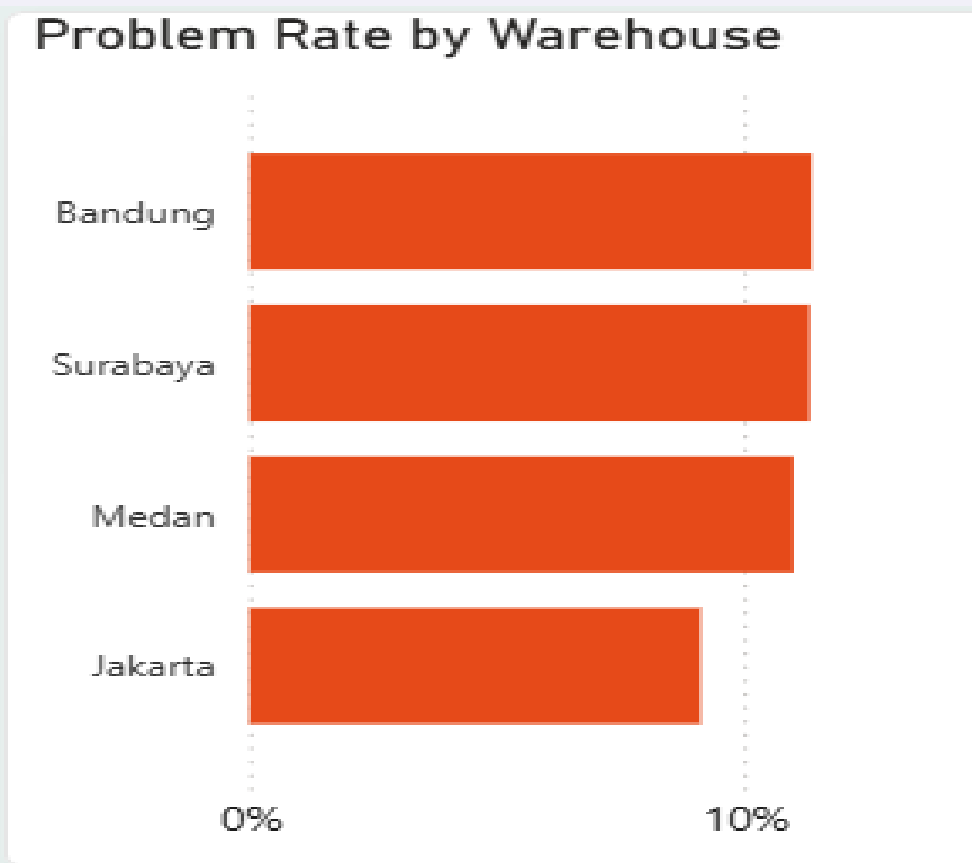
Operational Insight



What are the main reasons for product returns and cancellations?

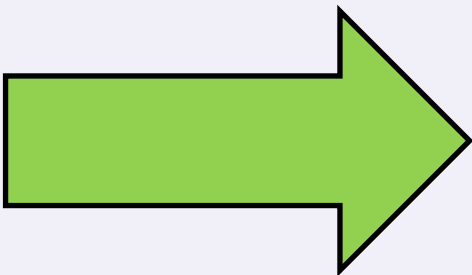
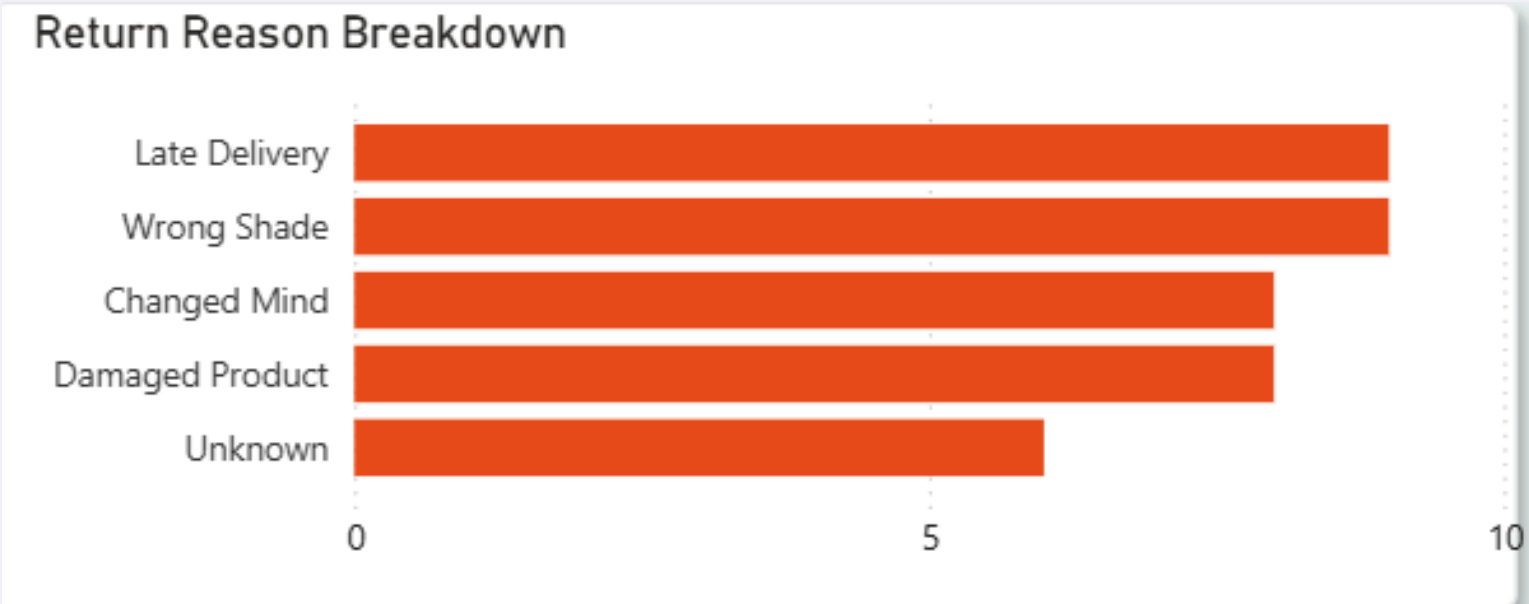
Operations suffer from a 5.3% cancellation rate and a 5.4% return rate. The leading reasons are “Wrong Shade” and “Changed Mind”, proving that friction is caused mainly by pre-purchase confusion, not only from post-purchase friction such as damaged goods.

Operational Insight



How fast are our deliveries, and which warehouses have the most problems?

The national average delivery time has crept up to 3.95 days. Surabaya warehouse struggles with the slowest fulfillment speed (4.07 days), while the Bandung warehouse records the highest overall problem transaction rate.



Return reason on Bandung Warehouse !

Bandung records the highest problem rate, with returns mainly driven by Late Delivery. This indicates the need to improve fulfillment speed.

RECOMMENDATIONS (DATA-DRIVEN DECISIONS)



Mitigate Channel Risks and Optimize Multi-Channel Profitability

- **Implement a Marketplace-to-D2C Funnel:** Use Shopee and TikTok Shop purely as low-barrier customer acquisition tools. Insert physical, high-value QR code vouchers inside marketplace delivery boxes offering exclusive discounts redeemable only on the official Website for their second purchase.
- **Optimize Margins on TikTok Shop:** Shift the TikTok strategy to mandate a strict minimum profit floor of 45% per live session and transition to exclusive bundles to prevent profit erosion from excessive discounting.



Maximize Margin Efficiency and Bundling Strategy

- **Hero Product Strategy on PRD-0006 :** Significantly increase its ad allocation, position it as the flagship item during Live Commerce sessions, and hardcode it as an automatic “recommended add-on” in the marketplace shopping cart.
- **Optimize Margin Structure of PRD-0004:** Restructure promotional strategies for PRD-0004 to halt margin dilution.
- **Monopolize the “Dry Skin” Segment:** Capitalize on the 41.28% dry skin consumer majority by designing high-value thematic bundles such as an “Anti-Crack Base Makeup Bundle” or a “Hydrating Beauty Set” to maximize the Average Order Value.
- **Scale Products (PRD-0023):** Scale PRD-0023 sales by pairing it in “Frequently Bought Together” bundles alongside Lipstick, and deploy targeted digital retargeting ads to past consumers.

RECOMMENDATIONS (DATA-DRIVEN DECISIONS)



Maximize Lifetime Value and Overcome the Satisfaction Deficit

- **Launch a 30-Day “CX Recovery” Program:** Address the critical 2.99 rating deficit immediately. Deploy customer service to conduct targeted follow-ups via WhatsApp for all High-Value customers who left a rating below 3 stars, offering sincere apologies and replacement guarantees.
- **Targeted Demographic Campaigns (Ages 25–44):** Pivot digital ad messaging to match the mature audience to maximize retention and lifetime value.
- **Create Gender-Specific Marketing Tracks:** Split the marketing funnel to capitalize on the male shopper. Launch specific Male-targeted ads.
- **Geospatial Strategy:** Focus on flash sales/free shipping in Volume Markets (Jawa Barat/Jawa Timur), premium upselling/same-day delivery in Value Markets (DKI Jakarta/Sumatra Utara).



Plug Profit Leaks and Optimize Fulfillment Efficiency

- **Deploy “Virtual Shade Guide”:** Directly combat the leading “Wrong Shade” return reason by enriching product pages with multi-lighting product photos, real-skin video swatches, and a comparative shade chart.
- **Implement a “Changed Mind” Friction Funnel:** Introduce an automated WhatsApp Order Confirmation for high-value Website orders and a strict 1-hour modification window post-checkout to give customers a brief grace period to correct their selection.
- **Initiate a Back-End Audit for Bandung & Surabaya:** Establish the Jakarta warehouse as the national SOP benchmark. Audit the Surabaya hub to reduce its 3.8-day delivery average, and audit the Bandung hub’s picking, packing, and return-handling accuracy due to its highest problem rate.

“

**Thank you for reading
through this case study**

–This project highlights how sales, customer, product, and operational data can be translated into actionable business strategies.